國立東華大學應用數學系

碩士論文

# 時空 skew-t 過程於門檻超標預測之延伸：AR(2) 時間先驗與多尺度樣條基底函數

***Extending the Spatio-Temporal Skew-t Process for Threshold Exceedance Prediction: AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions***

研究生：謝易軒
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中華民國 115 年 7 月

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# 摘要

在環境風險治理中，決策關心的通常不是平均濃度，而是未設站位置超越高門檻的機率。本文以 Morris et al. (2017) 的時空 skew-$t$ 過程 (STP) 爲基準，系統檢驗其可改進空間：在時間軸上，將潛在參數的 AR(1) 先驗擴充爲符合 Yule–Walker 標準化條件的 AR(2) 先驗；在空間軸上，於迴歸均值加入多尺度薄板樣條 (MRTS) 基底，以提升大尺度空間變異的表達能力。評估分爲模擬研究與實證分析。模擬研究涵蓋四組固定 AR(2) 生成設定與一組刻意設計的非定態隨機效應情境；實證分析使用 2005 年 7 月美國 AQS 臭氧資料與 CMAQ 共變數。模型比較以樣本外 Brier 分數爲核心指標，聚焦門檻超標預測表現。結果顯示，AR(2) 延伸在多數情境下未帶來穩定增益，且參數可識別性有限；MRTS 延伸僅在部分門檻與設定出現零散改善，未形成一致優勢。整體而言，兩項延伸的主要價值不在於取代 STP 基準，而在於將其擴展爲可由樣本外門檻分數進行資料驅動選擇的候選模型類。

**關鍵字**：時空 skew-$t$ 過程、門檻超標預測、AR(2) 先驗、多尺度薄板樣條、Brier 分數、臭氧

# Abstract

In environmental risk regulation, the inferential target is often not the field mean but the probability of exceeding a high threshold at unmonitored locations. We take the spatio-temporal skew-$t$ process (STP) of Morris et al. (2017) as baseline and probe where predictive improvement is still possible along two axes. On the temporal axis, we extend the latent AR(1) prior to a standardized AR(2) prior constrained by Yule–Walker moments. On the spatial axis, we augment the regression mean with multi-resolution thin-plate spline (MRTS) basis functions to capture large-scale spatial variation. Evaluation combines simulation and real-data analysis. The simulation study includes four fixed AR(2) generating settings and a deliberately constructed nonstationary random-effect scenario. The real-data study uses July 2005 U.S. AQS ozone observations with CMAQ as covariates. Models are compared by held-out Brier scores focused on threshold exceedance prediction. Across settings, AR(2) provides little stable gain and shows limited coefficient recoverability; MRTS yields occasional improvements but no uniform advantage. Thus the main contribution is to enlarge the STP baseline into a candidate model class selected by held-out threshold predictive performance.



**Keywords**: spatio-temporal skew-$t$ process; threshold exceedance prediction; AR(2) prior; multi-resolution thin-plate splines; Brier score; ozone

# Contents

# 1 Introduction

The need for accurate pollution-concentration estimates over broad spatial domains keeps growing. Epidemiological evidence has tied air quality to morbidity and mortality, and regulation has tightened. Direct measurement alone cannot meet this need: the cost and logistics of a monitoring network cap its spatial coverage and temporal resolution. This gap motivates statistical methods that extend a network's spatial reach with quantified confidence (U.S. Environmental Protection Agency, 2024). Ground-level ozone is the canonical regulatory case. The National Ambient Air Quality Standard (NAAQS) specifies the maximum concentration of ozone ($O_3$) allowed in outdoor air: a site complies only if the three-year average of its annual fourth-highest daily maximum eight-hour concentration stays at or below the current NAAQS of 0.070 parts per million (70 ppb), last revised in 2015. The Clean Air Act (CAA) requires the EPA to review it at least every five years. The standards for ozone are shown in Table 1. The policy-relevant target is thus not the mean field but its upper tail, namely the probability that an unmonitored location breaches a high threshold. We pursue two goals: spatial prediction at unmonitored sites, and mapping the probability of extreme exceedances across the domain.

Table 1: Ozone National Ambient Air Quality Standard

| Form of the Standard (parts per million, ppm) | **1997** | **2008** | **2015** |
|---|---|---|---|
| Annual $4^{\text{th}}$ highest daily max 8-hour average, averaged over three years | 0.080 | 0.075 | 0.070 |

A spatial model for ozone threshold exceedances warrants special consideration, since standard spatial methods are likely to perform poorly. Two features of the problem are decisive. First, likelihood-based spatial modeling typically assumes a Gaussian process, which is appropriate when interest lies in mean behavior. The Gaussian distribution is light-tailed and symmetric. Daily ozone maxima are neither. Their residuals are heavy-tailed and are well described by a skew-$t$ distribution (Morris, Reich, Emeric Thibaud, and Cooley, 2017). Second, our interest is in the dependence structure at high levels, which a covariance designed for mean behavior does not capture. A Gaussian process is asymptotically independent outside the degenerate case of perfect dependence, so the probability that two sites are jointly extreme decays to zero as the level increases. Nearby ozone monitors, however, tend to record high values on the same days, and this agreement weakens with distance (Morris, Reich, Emeric Thibaud, and Cooley, 2017). These considerations call for a heavy-tailed, asymmetric margin that retains asymptotic dependence. The skew-$t$ distribution provides both. It is a subasymptotic model, capturing the joint tail decay at high but finite levels rather than the limiting dependence of extreme events (Raphaël Huser and Wadsworth, 2022). This is the regime the ozone data occupy.

Our approach also departs from threshold models based on extreme-value distributions, a well-developed part of the field in which extreme events are defined as exceedances over a high threshold. Anthony C. Davison and Smith (1990) model univariate exceedances with the generalized Pareto distribution. Ledford and J. A. Tawn (1996) address the multivariate case through a censored likelihood that distinguishes the ways in which a threshold may be exceeded. Wadsworth and J. A. Tawn (2012) and E. Thibaud, Mutzner, and A. C. Davison (2013) carried these ideas to spatial extremes, fitting models by a censored pairwise likelihood (Padoan, Ribatet, and Sisson, 2010). R. Huser and A. C. Davison (2014) extended them to space-time data. Such methods are grounded in extreme-value theory and presume a threshold high enough for extremal models to be valid and for extrapolation beyond the observed range. They are also computationally demanding and, in practice, restricted to small datasets. Neither feature suits the present application. The $75$ ppb threshold corresponds to roughly the $0.92$ sample quantile of the daily ozone maxima (Morris, Reich, Emeric Thibaud, and Cooley, 2017), well short of the far tail that extremal models require. The network is also large, with $1{,}089$ monitoring stations. A complementary strategy retains the max-stable process while lightening its computational cost. Fully Bayesian inference for max-stable processes is cumbersome at scale. Morris, Reich, and Emeric Thibaud (2019) address this by truncating the spectral representation to a small set of empirical basis functions that carry the spatial dependence. Their construction renders inference feasible for large datasets and, as a by-product, dispenses with the assumption of stationarity.

In this paper we build on the spatial skew-$t$ process of Morris, Reich, Emeric Thibaud, and Cooley (2017). This process is not max-stable, yet it retains extremal dependence, and it is constructed by location-scale mixing of Gaussian processes. Their model has three components. A random spatial partition, redrawn each day, confines asymptotic dependence to nearby sites. A censored likelihood focuses inference on exceedances above a threshold. A transformed first-order autoregressive process carries dependence through time. We extend this baseline along two axes. The first is temporal. The autoregressive prior links each day only to the one before, and we add a further lag to capture longer-range dependence in the record. The second is spatial. The baseline assumes a stationary Matérn covariance, which is restrictive over a continental network. We therefore introduce the multi-resolution thin-plate spline basis of Tzeng and Huang (2018) as fixed-effect covariates in the mean, following the basis-function idea reviewed above, so as to relax stationarity at little additional cost. Each axis is a candidate extension rather than a replacement of the baseline, and together they are intended to improve prediction of threshold exceedances.

The remainder of the thesis is organized as follows. Section 2 reviews the spatial skew-$t$ process of Morris, Reich, Emeric Thibaud, and Cooley (2017), develops the two extensions in turn (the standardized AR(2) prior on the latent dynamics and the MRTS basis functions in the mean), and assembles them into a single hierarchical model together with the MCMC algorithm used for inference. Section 3 evaluates each extension on synthetic data, describing the generating designs, the cross-validation scheme, and the held-out Brier score, and reports the comparisons for the AR(2) and MRTS extensions separately. Section 4 applies the models to the ozone monitoring data, comparing the baseline against the two extensions in their ability to predict threshold exceedances. Section 5 concludes and outlines directions for further work. Technical derivations for the AR(2) prior are collected in the appendices.

# 2 Methodology

In this section, we first introduce the spatio-temporal skew-$t$ process (STP) proposed by Morris, Reich, Emeric Thibaud, and Cooley (2017), and then develop two extensions of it. The first extends the autoregressive structure on the Gaussian-copula margins of the latent processes from AR(1) to AR(2). The second augments the regression component with multi-resolution thin-plate spline (MRTS) basis functions to capture non-stationary mean structure. Lastly, we present the full hierarchical model together with the MCMC algorithm used for inference.

## 2.1 The spatio-temporal skew-$t$ model

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \ldots, n_t\}$. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), we model

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} \;+\; \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \tag{1}$$

where $\mathbf{X}_t(\mathbf{s})$ is a $p$-vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient vector, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic Matérn correlation

$$\rho(h) \;=\; (1-\gamma)\, I(h=0) \;+\; \gamma\, \frac{1}{\Gamma(\nu)\, 2^{\nu-1}} \left(\sqrt{2\nu}\,\frac{h}{\varphi}\right)^{\nu} K_{\nu}\!\left(\sqrt{2\nu}\,\frac{h}{\varphi}\right), \tag{2}$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\gamma \in [0,1]$ is the proportion of variance attributed to the spatial component, $K_\nu$ is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ is the Euclidean distance between sites $\mathbf{s}_1$ and $\mathbf{s}_2$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \ldots, Y_t(\mathbf{s}_n)]^\top$ follows an $n$-dimensional skew-$t$ distribution; consequently (1) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

**Censoring.** Because the EPA compliance criterion focuses on exceedances of a high level $L$, the model fits only the upper tail by censoring the data at a pre-specified threshold $T \le L$. Writing the partially censored response $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\widetilde{\mathbf{Y}}_t = [\widetilde{Y}_t(\mathbf{s}_1), \ldots, \widetilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below $T$ are imputed inside the MCMC.

**Random Partition.** To remove the long-range asymptotic dependence that arises when a single scalar $z$ and $\sigma^2$ are shared across the entire domain, the model uses a daily random partition. For each day $t$, let $\mathbf{w}_{t1}, \ldots, \mathbf{w}_{tK}$ denote $K$ spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} \;=\; \{\mathbf{s} \in \mathcal{D} : k = \arg\min_{\ell} \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \qquad k = 1, \ldots, K. \tag{3}$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \overset{iid}{\sim} N(0,1)$ and $\sigma_{tk}^2 \overset{iid}{\sim} \mathrm{IG}(a/2, b/2)$. Within each subregion the model retains the skew-$t$ form, while sites in different partitions are not asymptotically dependent.

**Temporal Dependence.** Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents $z_{tk}$, and the scale parameters $\sigma_{tk}^2$. To preserve the marginal skew-$t$ distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform (PIT):

$$w_{tki}^{*} = \Phi^{-1}\left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)}\right], \quad i = 1, 2, \tag{4}$$

$$z_{tk}^{*} = \Phi^{-1}\{\mathrm{HN}(z_{tk})\}, \tag{5}$$

$$\sigma^{2*}_{tk} = \Phi^{-1}\{\mathrm{IG}(\sigma^2_{tk})\}, \tag{6}$$

where $\Phi$ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}^*_{tk}, z^*_{tk}, \sigma^{2*}_{tk}$ has a standard normal marginal distribution, and Morris, Reich, Emeric Thibaud, and Cooley (2017) place a stationary AR(1) time series on each of them:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k} \sim N_2\big(\phi_w\, \mathbf{w}^*_{t-1,k},\ (1-\phi_w^2)\, \mathbf{I}_2\big)\,, \tag{7}$$
$$z^*_{tk} \mid z^*_{t-1,k} \sim N\big(\phi_z\, z^*_{t-1,k},\ 1-\phi_z^2\big)\,, \tag{8}$$
$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k} \sim N\big(\phi_\sigma\, \sigma^{2*}_{t-1,k},\ 1-\phi_\sigma^2\big)\,, \tag{9}$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$, and initial values $\mathbf{w}^*_{1k} \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z^*_{1k} \sim N(0,1)$, $\sigma^{2*}_{1k} \sim N(0,1)$. Transforming back through (4)–(6) gives $\mathbf{w}_{tk} \sim \mathrm{Unif}(\mathcal{D})$, $z_{tk} \sim \mathrm{HN}(0,1)$, and $\sigma^2_{tk} \sim \mathrm{IG}(a/2, b/2)$ at every $t$, so that the spatial skew-$t$ structure of (1) is preserved at each day.

## 2.2 Extremal dependence

For exceedance prediction, what matters is how strongly the skew-$t$ process couples extreme values at two sites, and how that coupling changes with their separation. One measure of this extremal dependence is the $\chi$ statistic (Coles, Heffernan, and J. Tawn, 1999). For a stationary and isotropic spatial process, the $\chi$ statistic for two locations separated by distance $h$ is

$$\chi(h) = \lim_{c \to c^*} \Pr[Y(\mathbf{s}+h) > c \mid Y(\mathbf{s}) > c], \tag{10}$$

where $c^*$ is the upper limit of the support of $Y$; for the skew-$t$ distribution $c^* = \infty$. If $\chi(h) = 0$, then observations are asymptotically independent at distance $h$. For Gaussian processes, $\chi(h) = 0$ regardless of the distance $h$, so they are not suitable for modeling asymptotically dependent extremes. Unlike the Gaussian process, the skew-$t$ process is asymptotically dependent. However, one problem with the spatial skew-$t$ process is that $\lim_{h\to\infty} \chi(h) > 0$. This occurs because all observations, both near and far, share the same $z$ and $\sigma$ terms. Such long-range dependence is undesirable over large domains, where only local dependence is expected. Morris, Reich, Emeric Thibaud, and Cooley (2017) avoid this problem through a random partitioning mechanism, described in detail in Section 2.1.

## 2.3 Extension to AR(2) time-varying coefficients

In this section, we generalize the AR(1) prior (7)–(9) on the transformed latents to a stationary AR(2) prior.

**The AR(2) prior.** Writing $\{Y_t\}$ for any scalar component of $\mathbf{w}^*_{tk}$, or for $z^*_{tk}$ or $\sigma^{2*}_{tk}$ at a fixed knot $k$, the prior is the stationary AR(2) recursion

$$Y_t \;=\; \phi_1 Y_{t-1} \;+\; \phi_2 Y_{t-2} \;+\; \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} N(0,\sigma^2_\epsilon), \qquad t \geq 3, \tag{11}$$

with $(\phi_1, \phi_2)$ confined to the stationarity triangle

$$|\phi_2| < 1, \qquad \phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1. \tag{12}$$

Because the inverse PIT (4)–(6) returns the half-normal, inverse-gamma, and uniform target margins only when $\gamma_0 := \mathrm{Var}(Y_t) = 1$, the innovation variance is not free: solving the Yule–Walker equations under $\gamma_0 = 1$ (Appendix A.1) fixes the standardized autocovariances

$$\gamma_1 \;=\; \frac{\phi_1}{1-\phi_2}, \qquad \gamma_2 \;=\; \phi_1\,\gamma_1 + \phi_2, \tag{13}$$

$$\sigma^2_\epsilon \;=\; 1 - \phi_1\,\gamma_1 - \phi_2\,\gamma_2. \tag{14}$$

Setting $\phi_2 = 0$ gives $\gamma_1 = \phi_1$ and $\sigma^2_\epsilon = 1 - \phi_1^2$, the AR(1) form of (7)–(9), so the AR(2) prior strictly nests the prior of Morris, Reich, Emeric Thibaud, and Cooley (2017).

**Stationary initial pair.** The recursion (11) needs two seeds, and independent $N(0,1)$ seeds would leave $\mathrm{Var}(Y_t) \neq 1$ over the first days and break the margin. We therefore draw the initial pair from the stationary bivariate Gaussian

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \;\sim\; N_2\!\left( \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \tag{15}$$

so that $\mathrm{Var}(Y_t) = 1$ holds from $t = 1$ onward.

**Application to the three transformed parameters.** Mirroring the AR(1) block (7)–(9), the AR(2) prior (11) with initial condition (15) replaces it component-wise:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k}, \mathbf{w}^*_{t-2,k} \sim N_2\left(\phi_1^{(w)}\, \mathbf{w}^*_{t-1,k} + \phi_2^{(w)}\, \mathbf{w}^*_{t-2,k},\ \sigma^2_{\epsilon,w}\, \mathbf{I}_2\right), \tag{16}$$

$$z^*_{tk} \mid z^*_{t-1,k}, z^*_{t-2,k} \sim N\left(\phi_1^{(z)}\, z^*_{t-1,k} + \phi_2^{(z)}\, z^*_{t-2,k},\ \sigma^2_{\epsilon,z}\right), \tag{17}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k}, \sigma^{2*}_{t-2,k} \sim N\left(\phi_1^{(\sigma)}\, \sigma^{2*}_{t-1,k} + \phi_2^{(\sigma)}\, \sigma^{2*}_{t-2,k},\ \sigma^2_{\epsilon,\sigma}\right), \tag{18}$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies in the stationarity triangle (12), the innovation variances follow from (13)–(14), and the initial pairs follow (15) component-wise. After the inverse PIT (4)–(6), the back-transformed $\mathbf{w}_{tk}, z_{tk}, \sigma^2_{tk}$ keep the same marginals as the AR(1) baseline, so the spatial skew-$t$ model (1) is preserved at every $t$. The Gibbs-within-MH full conditional that updates each $Y_t$ now carries three prior factors (self, lag-1, lag-2); its explicit form and the Metropolis–Hastings acceptance ratio are given in Appendix A.2.

## 2.4 Incorporating MRTS covariates

Tzeng and Huang (2018) proposed the multi-resolution thin-plate spline (MRTS) basis function, a basis whose functions are ordered from coarse to increasingly fine resolution. This ordering lets a model add spatial detail one resolution at a time; we define the basis below on a 2-dimensional spatial domain. Consider $n$ distinct locations, $\mathbf{s}_1, \cdots, \mathbf{s}_n$, where $\mathbf{s}_i = (x_{i1}, x_{i2})^\top \in \mathcal{R}^2$ for $i = 1, \ldots, n$, and let

$$\boldsymbol{X} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{pmatrix} \tag{19}$$

. Then the MRTS basis functions $f_k(\mathbf{s})$ at a location $\mathbf{s} = (x_1, x_2)^\top$ are defined as

$$f_k(\mathbf{s}) = \begin{cases} 1, & k = 1, \\ x_{k-1}, & k = 2, 3, \\ \lambda_{k-3}^{-1}\left\{\boldsymbol{\phi}(\mathbf{s}) - \boldsymbol{\Phi}\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}\right\}^\top \mathbf{v}_{k-3}, & k = 4, \ldots, n, \end{cases} \tag{20}$$

where $\boldsymbol{x} = (1, x_1, x_2)^\top$, $\boldsymbol{\phi}(\mathbf{s}) = (\phi_1(\mathbf{s}), \ldots, \phi_n(\mathbf{s}))^\top$ with

$$\phi_i(\mathbf{s}) \;=\; \frac{1}{8\pi}\, \|\mathbf{s} - \mathbf{s}_i\|^2 \,\log\bigl(\|\mathbf{s} - \mathbf{s}_i\|\bigr), \qquad i = 1, \ldots, n, \tag{21}$$

and $\mathbf{\Phi}$ is the $n \times n$ matrix with $(i, j)$-th element $\phi_j(\mathbf{s}_i)$, $\mathbf{v}_k$ is the $k$-th column of $\mathbf{V}$, and $\mathbf{V}\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\,\mathbf{V}^\top$ is the eigen-decomposition of $\mathbf{Q}\,\mathbf{\Phi}\,\mathbf{Q}$ with $\lambda_1 \geq \cdots \geq \lambda_n$ and $\mathbf{Q} = \mathbf{I} - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$.

The MRTS basis enters the STP through the regression component. Fix the number of basis functions $K_{\mathrm{MRTS}} \geq 1$ and let

$$\mathbf{f}(\mathbf{s}) \;=\; \bigl(f_1(\mathbf{s}),\, f_2(\mathbf{s}),\, \ldots,\, f_{K_{\mathrm{MRTS}}}(\mathbf{s})\bigr)^\top \tag{22}$$

collect the first $K_{\mathrm{MRTS}}$ basis functions defined in (20). Then the day-$t$ covariate vector $\mathbf{X}_t(\mathbf{s})$ in (1) is replaced by

$$\begin{aligned} \mathbf{X}_t^*(\mathbf{s}) &= \bigl[\mathbf{f}(\mathbf{s})^\top,\; \mathbf{X}_t(\mathbf{s})^\top\bigr]^\top \\ &= \bigl[f_1(\mathbf{s}),\; f_2(\mathbf{s}),\; \ldots,\; f_{K_{\mathrm{MRTS}}}(\mathbf{s}),\; \mathrm{X}_1(\mathbf{s}, t),\; \ldots,\; \mathrm{X}_p(\mathbf{s}, t)\bigr]^\top, \end{aligned} \tag{23}$$

where $\mathrm{X}_j(\mathbf{s}, t)$ represents the $j$-th exogenous covariate of $\mathbf{X}_t(\mathbf{s})$. The model in (1) can then be expressed as

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* \;+\; \lambda\,\sigma_t(\mathbf{s})\,|z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\,v_t(\mathbf{s}), \tag{24}$$

which is identical to the STP model (1) on all components other than the mean, with $\boldsymbol{\beta}^* \in \mathcal{R}^{K_{\mathrm{MRTS}}+p}$. Because the MRTS basis functions depend only on location, each site $\mathbf{s}$ shares the same ordered set of basis values $\mathbf{f}(\mathbf{s})$ at every day $t$. The number of basis functions $K_{\mathrm{MRTS}}$ is the practical tuning parameter: Section 3.1 compares $K_{\mathrm{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$ on simulated data of different models.

## 2.5 Hierarchical model

Following Morris, Reich, Emeric Thibaud, and Cooley (2017), we write the spatial skew-$t$ threshold-exceedance model as a Bayesian hierarchical model, fit by Markov chain Monte Carlo (MCMC). The data are partially censored skew-$t$ observations: at each MCMC iteration we impute the values below the threshold $T$, so that estimation proceeds on the completed $\mathbf{Y}$. Conditional on the partition latents $z_{tk}(\mathbf{s})$, $\sigma^2_{tk}(\mathbf{s})$ and the subregions $P_{tk}$, $\mathbf{Y}$ is multivariate Gaussian. The knot locations $\mathbf{w}$ that define the partition are themselves unknown and random. Their number of knots $K$, by contrast, we do not also make random, since that is computationally onerous; instead we fix it at several values and select among them by held-out Brier scores (Section 3.2). The hierarchical model is

$$
\begin{aligned}
Y_t(\mathbf{s}) \mid z_t(\mathbf{s}), \sigma_t^2(\mathbf{s}), P_{tk}, \Theta &= \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} + \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| + \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \qquad (25)\\
z_t(\mathbf{s}) &= z_{tk} \text{ if } \mathbf{s} \in P_{tk},\\
\sigma_t^2(\mathbf{s}) &= \sigma_{tk}^2 \text{ if } \mathbf{s} \in P_{tk},\\
\lambda &\sim N(0, \sigma_\lambda^2),\\
v_t(\mathbf{s}) \mid \Theta &\sim \text{Matérn}\ (\mathbf{0}, \boldsymbol{\Sigma}),
\end{aligned}
$$

where $\Theta = \{\varphi, \nu, \gamma, \lambda, \boldsymbol{\beta}\}$ and $\boldsymbol{\Sigma}$ is the Matérn covariance of (2). The priors on $\mathbf{w}_{tk}^*$, $z_{tk}^*$ and $\sigma_{tk}^{2*}$ follow (7)–(9) for the baseline, or (16)–(18) under our AR(2) extension. Equation (25) is the baseline of Morris, Reich, Emeric Thibaud, and Cooley (2017); our contributions are the AR(2) priors above and the MRTS mean term $\mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^*$ of (24), which replace the latent dynamics and the mean while leaving the censoring layer and Matérn covariance unchanged. Predictions at new sites use Bayesian kriging on the multivariate normal full conditional of $\mathbf{Y}_t$.

# 3 Simulation study

In this section, we present the results from a simulation study to investigate how the AR(2) time-varying structure on latent variables and the number of thin-plate spline basis functions impact the accuracy of predictions made by the model and to compare with Gaussian methods and methods without the proposed extensions.

## 3.1 Design

All simulation designs generate data from the hierarchical model of Section 2.5 on $n_s = 144$ sites, drawn uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ days ($n_t = 200$ in the time-block forecast arm of the persistence experiment below); the single exception is the pair of nonlinear-mean designs of the MRTS experiment, which replace the skew-$t$ error process with a Gaussian one. In the data generation the regression mean is held constant, $\mathbf{X}^\top \boldsymbol{\beta} = 10$ (the MRTS experiment below adds a mean-zero non-stationary random effect to one design, which leaves the regression mean at $10$, and a fixed nonlinear mean surface to its two Gaussian-error designs), while every fitted model includes an intercept and a first-order spatial trend. For each design we simulate $10$ data sets ($3$ for the transform-mix pilot of the MRTS experiment), each split into $100$ training sites and $44$ withheld test sites, and run every MCMC chain for $20,000$ iterations with a burn-in period of $10,000$ iterations. Throughout, $q(0.80)$ denotes the $80$th sample quantile of the data, and the five baseline analysis models are

1. Gaussian marginal, $K = 1$ knot;
2. Skew-$t$ marginal, $K = 1$ knot, no thresholding;
3. Symmetric-$t$ marginal, $K = 1$ knot, threshold $T = q(0.80)$;
4. Skew-$t$ marginal, $K = 5$ knots, no thresholding;
5. Symmetric-$t$ marginal, $K = 5$ knots, threshold $T = q(0.80)$.

**Experiment for AR(2) extension.** This experiment embeds a fixed AR(2) mechanism into the skew-$t$ generating process, so the AR(2) analysis model can be tested against data with a known autoregressive structure. All four generating designs are skew-$t$ with $\lambda = 3$, with the latent $\sigma^2$, $\mathbf{w}$, and $z$ processes evolving through a fixed AR(2) recursion (16)–(18) whose coefficients are constant across time:

1. $K = 1$ knot, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
2. $K = 1$ knot, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
3. $K = 5$ knots, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
4. $K = 5$ knots, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$.

Both coefficient pairs lie inside the stationarity triangle (12). Each data set is fit by nine analysis models: the five baseline models above, together with four temporal variants: skew-$t$ models with AR(1) and AR(2) time-varying coefficients (following (16)–(18)) at $K = 1$ and $K = 5$ knots.

**Experiment for persistent and long-memory dependence.** The four fixed-AR(2) designs above all have short memory: their autocorrelations decay geometrically within a few days. This experiment probes the two persistence regimes in which the autoregressive order could plausibly matter. Both designs are skew-$t$ with $K = 1$ knot and $\lambda = 3$, with the dependence imposed on the same transformed latent processes as above:

1. near-unit-root AR(2), $(\phi_1, \phi_2) = (0.15, 0.80)$: the pair lies inside the stationarity triangle (12), but the dominant root of the characteristic polynomial is $0.973$, so shocks persist for roughly a hundred days rather than a few; the dependence is dominated by the second lag ($\phi_2 \gg \phi_1$), the one feature an AR(1) model cannot represent;

2. long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$: the fractionally differenced process $(1 - B)^d \xi_t = \epsilon_t$, where $B$ is the backshift operator (Granger and Joyeux, 1980; Hosking, 1981), has hyperbolically decaying, non-summable autocorrelations, $\rho(h) \sim h^{2d-1}$, so no finite-order autoregression reproduces it and both the AR(1) and the AR(2) analysis models are misspecified by construction; sample paths are generated exactly by circulant embedding (Davies and Harte, 1987).

Each design is evaluated in two arms. The *spatial hold-out arm* repeats the cross-validation design above ($n_t = 50$; 10 data sets; 100 training and 44 test sites), fitting the Gaussian model and the skew-$t$ $K = 1$ models with AR(1) and AR(2) structure. The *time-block forecast arm* generates longer records ($n_t = 200$) on the same 144 sites, fits each model to days 1–50 at all sites, and predicts days 51–65, i.e., forecast leads $h = 1, \dots, 15$; the fitted models are the skew-$t$ $K = 1$ variants with the latent processes i.i.d. in time, with AR(1) structure, and with AR(2) structure. Fits failing a pre-specified consistency check (recovery of $\lambda$ and the intercept, and a bounded predictive spread) are re-run with a fresh seed and excluded if they fail again; scores are compared on the first 10 data sets whose fits pass under all three models.

**Experiment for MRTS extension.** This experiment enriches the regression component with the MRTS basis functions $f_1(\mathbf{s}), \dots, f_{K_{\text{MRTS}}}(\mathbf{s})$ of Section 2.4. We use four generating designs; the first two share the skew-$t$ core:

1. Skew-$t$ , $K = 5$ knots, $\lambda = 3$, with the constant mean $\mathbf{X}^\top \boldsymbol{\beta} = 10$;

2. Skew-$t$ , $K = 1$ knot, $\lambda = 3$, with an additive non-stationary spatial random effect $u_t(\mathbf{s})$ (specified below).

The second design moves the non-stationarity into the dependence, to expose the structural limit of a fixed-effect mean basis. It adds a mean-zero, per-day random effect built from the cosine basis, on $[0, 10]^2$ domain:

- Two fixed cosine fields with centres $\mathbf{c}_1 = (0, 10)^\top$, $\mathbf{c}_2 = (7.5, 2.5)^\top$ and native wavenumbers $\kappa_1 = \pi/10$, $\kappa_2 = \pi/5$,

$$\phi_j(\mathbf{s}) = \cos\big(\kappa_j \, \|\mathbf{s} - \mathbf{c}_j\|\big), \qquad j = 1, 2. \tag{26}$$

- The random effect superposes them with independent, daily redrawn Gaussian weights,

$$u_t(\mathbf{s}) = \sum_{j=1}^{2} \xi_{tj} \, \phi_j(\mathbf{s}), \qquad \xi_{tj} \overset{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3), \tag{27}$$

  independently across $j$ and across days $t$.

The response adds $u_t$ to the 1-knot skew-$t$ model,

$$Y_t(\mathbf{s}) = 10 + \lambda \, \sigma_t(\mathbf{s}) \, |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) \, v_t(\mathbf{s}) + u_t(\mathbf{s}). \tag{28}$$

Marginalising over $\xi$ gives a zero-mean random effect whose pointwise variance $\sum_j \tau_j^2 \, \phi_j(\mathbf{s})^2$ varies with location (Appendix A.3), so the dependence is non-stationary while the marginal mean stays constant at $10$.

These two designs leave the time-averaged mean without exploitable structure: the five-knot design has a constant mean by construction, and the random effect $u_t$ is mean-zero. To probe the complementary regime, in which the non-stationarity lives entirely in the first moment, two further designs carry a fixed nonlinear mean surface and drop the skew-$t$ generator altogether,

$$Y_t(\mathbf{s}) \;=\; 10 \;+\; g(\mathbf{s}) \;+\; \sigma_g \, \varepsilon_t(\mathbf{s}), \qquad \sigma_g = 4, \tag{29}$$

where the errors $\varepsilon_t$ are standard Gaussian processes with the Matérn correlation (2) at the generating values used throughout ($\gamma = 0.9$, $\nu = 1/2$, $\varphi = 1$), drawn independently across days. The marginal law is therefore Gaussian, so every $t$-family analysis model is misspecified in its error law and the mean channel is tested without the assistance of a correctly specified margin. Each raw surface $\tilde{g}$ is orthogonalized against the baseline design $[1, s_1, s_2]$ and standardized to a spatial standard deviation of $11$, so the $K_{\text{MRTS}} = 0$ design spans none of the resulting $g$.

The two surfaces are:

1. arc front: $\tilde{g}(\mathbf{s}) = \tanh\{\|\mathbf{s} - (2,2)^\top\| - 4.5\}$, a smoothed circular step whose transition band of width $\approx 2$ is the steepest smooth localized structure the $100$-site training density can resolve;

2. transform mix: $\tilde{g}(\mathbf{s}) = \mathrm{std}\{\log(1+s_1)\} + \mathrm{std}\{e^{-s_1/2}\} + \mathrm{std}\{s_1/(1+s_2)\}$, with $\mathrm{std}\{\cdot\}$ standardizing each term to unit spatial standard deviation: the nonlinear regression terms an applied modeler actually writes down, whose curvature concentrates near the domain boundary, exactly where the uniform site draw is sparsest.

Each of the five baseline models is fit across a grid of MRTS basis sizes, $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ recovers the no-MRTS baseline; on the arc-front design the grid is extended by $\{30, 40, 50\}$, since the front is resolved only by a fine basis. The arc-front design is run at the full scale of $10$ data sets, while the transform-mix design is a pilot, fitted on $3$ data sets at $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$, and its results are read as indicative.

## 3.2 Cross validation

Models are compared using cross validation, with $100$ sites used as training sites to fit each setting and $44$ sites withheld for evaluating predictions on each replicate. Because one of the primary goals of this model is to predict exceedances over a high level, we use Brier scores to compare the models (Gneiting and Raftery, 2007). For an exceedance level $L$, the Brier score is

$$\mathrm{BS}(L) \;=\; \left\{\, I(y > L) \;-\; \widehat{P}(Y > L) \,\right\}^2, \tag{30}$$

where $y$ is the held-out observation, $I(y > L)$ is the exceedance indicator, and $\widehat{P}(Y > L)$ is the posterior predictive probability of exceeding $L$. We average the Brier scores over all test sites and days; lower values indicate better exceedance prediction.

**Time-block forecast validation.** The site-splitting scheme above scores interpolation: when a withheld site is predicted, every day of the record is observed at the training sites. The time-block arm of the persistence experiment scores temporal extrapolation instead: all $144$ sites are observed for days $1$–$50$, and predictions are evaluated at the same sites on the following

15 days. For each forecast lead $h$ the Brier score (30) is computed with $L$ set to the empirical $0.95$ quantile of the validation block (the same threshold for every model), averaged over sites, and then over data sets; models are compared pairwise within each data set.

### 3.3 Results on AR(2) comparison

In general, we find that the methods whose marginal matches the generating truth give the largest improvement over the Gaussian model, while a mismatched model can perform worse than Gaussian, particularly in the far tail (Figure 1). The preferred knot count tracks the data-generating truth rather than the strength of the temporal dependence: the one-knot models are best when the data are generated with one knot, and the five-knot models are best when the data are generated with five knots, with the AR(2) coefficients shifting the magnitude of the scores but not their ordering.

Adding AR(2) (or AR(1)) temporal structure does not help. Even on data explicitly generated with an AR(2) recursion (the most favorable possible setting for the temporal extension), the temporal curves sit essentially on top of their non-temporal counterparts at every quantile and in every setting. Predictive accuracy is governed by the flexibility of the marginal and the spatial partition, namely the skew-$t$ margin and the number of knots, rather than by the autoregressive order of the latent coefficients.

The recovery of the AR(2) coefficients tells the same story (Table 2). Only the scale process is recovered with any fidelity, and only under strong dependence at a matched one-knot fit; the skewness process is recovered only partially and the knot-location process is essentially noise. Recovery deteriorates as the knot count grows and as the dependence weakens, so even on AR(2)-generated data the temporal signal is only weakly identified. In summary, the AR(2) structure should be treated as a candidate option rather than a default improvement.

### 3.4 Results under persistent and long-memory dependence

In the spatial hold-out arm the two persistence designs repeat, in sharper form, the null pattern of the fixed-AR(2) settings (Figure 2). Both temporal variants improve on the Gaussian model by only one to two percent, and they are indistinguishable from each other: at the $0.95$ threshold quantile the relative Brier scores of the AR(2) and AR(1) fits are $0.986$ versus $0.988$ under the near-unit-root design and $0.9837$ versus $0.9838$ under long memory, and no quantile separates the two by more than a fraction of a percent. This insensitivity is not for want of signal. Under the near-unit-root design the AR(2) fit recovers the generating coefficients well,

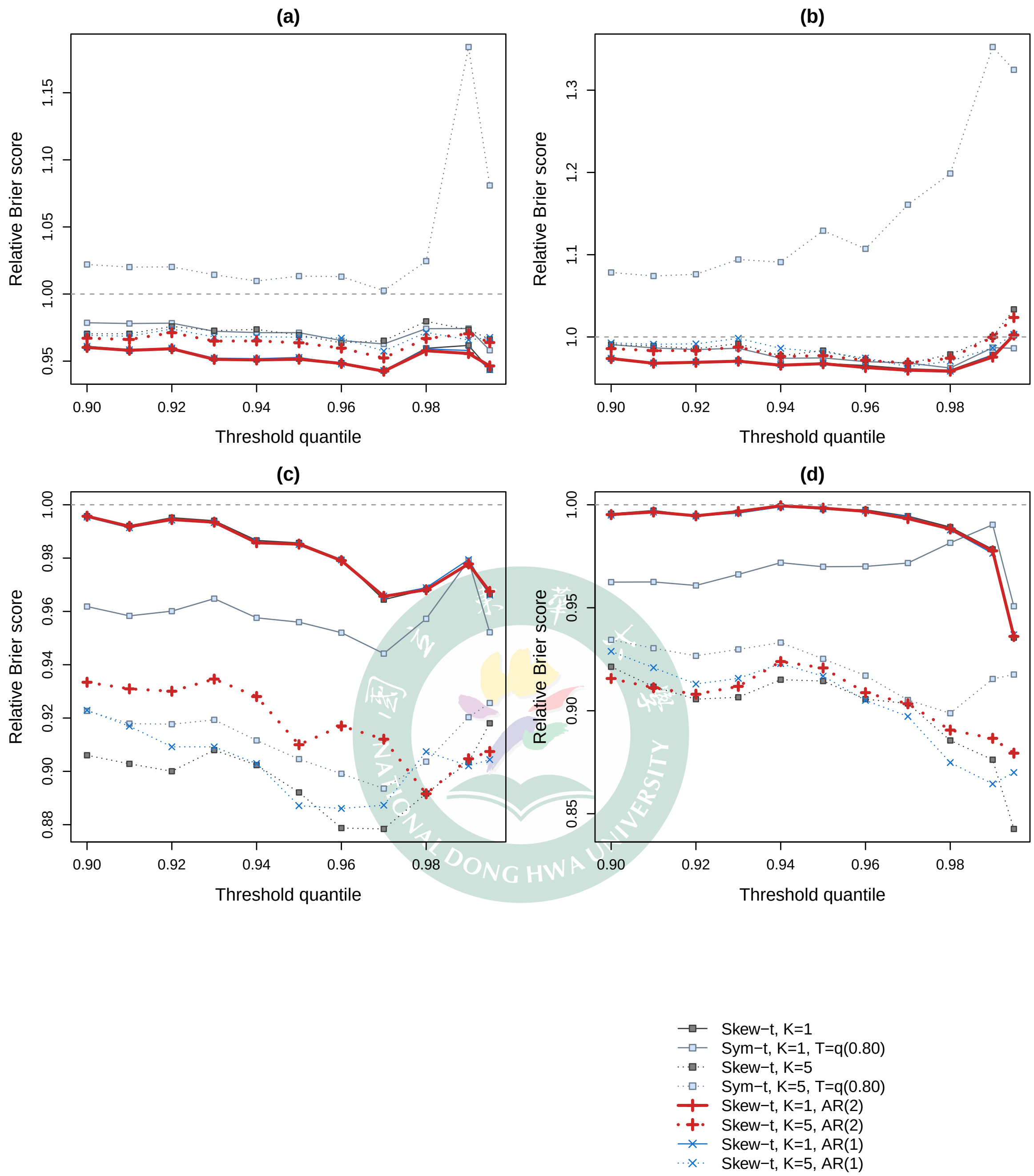



Figure 1: Relative Brier score versus the Gaussian model by threshold quantile (Gaussian is the reference $\equiv 1$, shown only as the dashed line; values below it are better). Data-generating settings: (a) skew-$t$ ($K = 1$, $\lambda = 3$), AR(2): $\phi_1 = 0.8$, $\phi_2 = -0.35$; (b) skew-$t$ ($K = 1$, $\lambda = 3$), AR(2): $\phi_1 = 0.12$, $\phi_2 = -0.05$; (c) skew-$t$ ($K = 5$, $\lambda = 3$), AR(2): $\phi_1 = 0.8$, $\phi_2 = -0.35$; (d) skew-$t$ ($K = 5$, $\lambda = 3$), AR(2): $\phi_1 = 0.12$, $\phi_2 = -0.05$.

Table 2: Posterior mean (SD) of the recovered AR(2) coefficients in the four fixed-AR(2) settings. Gen $K$ is the data-generating knot count and Fit $K$ the analysis-model knot count; $\hat{\phi}^{(\sigma)}$, $\hat{\phi}^{(z)}$, $\hat{\phi}^{(w)}$ are the estimated AR(2) coefficients of the scale, skewness, and knot-location processes ((16)–(18)).

| Settings | | Fit | $\hat{\phi}^{(\sigma)}$ | | $\hat{\phi}^{(z)}$ | | $\hat{\phi}^{(w)}$ | |
|---|---|---|---|---|---|---|---|---|
| $(\phi_1, \phi_2)$ | $K$ | $K$ | $\hat{\phi}_1$ | $\hat{\phi}_2$ | $\hat{\phi}_1$ | $\hat{\phi}_2$ | $\hat{\phi}_1$ | $\hat{\phi}_2$ |
| $(0.80, -0.35)$ | 1 | 1 | 0.803(0.140) | -0.363(0.137) | 0.626(0.114) | -0.225(0.157) | - | - |
| $(0.80, -0.35)$ | 1 | 5 | 0.871(0.142) | -0.438(0.164) | 0.472(0.284) | -0.270(0.162) | -0.129(0.377) | 0.062(0.257) |
| $(0.12, -0.05)$ | 1 | 1 | 0.083(0.122) | -0.019(0.150) | 0.186(0.137) | 0.084(0.226) | - | - |
| $(0.12, -0.05)$ | 1 | 5 | 0.095(0.141) | -0.050(0.220) | 0.050(0.088) | 0.051(0.142) | 0.002(0.215) | 0.179(0.259) |
| $(0.80, -0.35)$ | 5 | 1 | 0.512(0.190) | 0.043(0.245) | 0.434(0.185) | -0.108(0.128) | - | - |
| $(0.80, -0.35)$ | 5 | 5 | 0.169(0.115) | -0.023(0.118) | 0.074(0.093) | -0.006(0.090) | -0.019(0.206) | 0.094(0.135) |
| $(0.12, -0.05)$ | 5 | 1 | 0.285(0.228) | 0.161(0.164) | 0.163(0.122) | -0.063(0.122) | - | - |
| $(0.12, -0.05)$ | 5 | 5 | -0.001(0.072) | -0.017(0.137) | 0.010(0.075) | -0.010(0.067) | -0.017(0.193) | 0.115(0.190) |

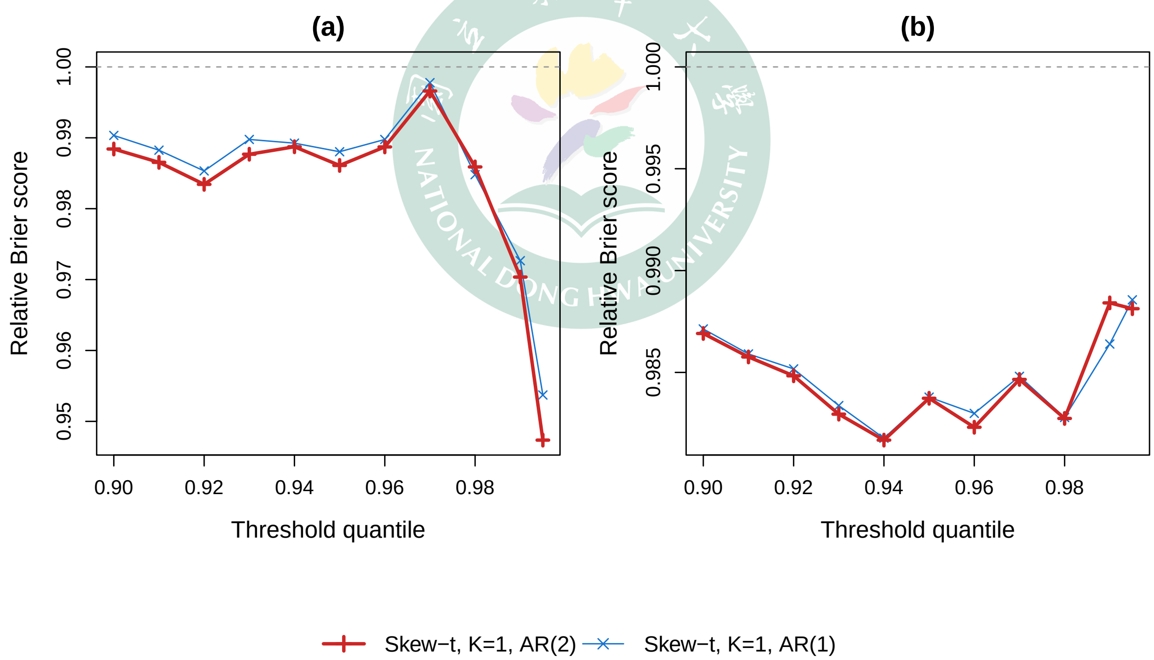



Figure 2: Spatial hold-out arm of the persistence experiment: relative Brier score versus the Gaussian model by threshold quantile (Gaussian is the reference $\equiv 1$, shown as the dashed line; values below it are better). Data-generating settings: (a) near-unit-root AR(2), $(\phi_1, \phi_2) = (0.15, 0.80)$; (b) long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$. Both are skew-$t$ ($K = 1$, $\lambda = 3$).

with posterior means $(\hat{\phi}_1, \hat{\phi}_2) = (0.11, 0.74)$ for the scale process and $(0.17, 0.73)$ for the skewness process against the truth $(0.15, 0.80)$, while the AR(1) fit compresses the same persistence into a single lag ($\hat{\phi}_1 = 0.33$ and $0.53$, respectively); under long memory the AR(2) fit likewise picks up a clearly positive second lag ($\hat{\phi}_2 \approx 0.20$ on the scale process), the signature of the best finite-order approximation to the hyperbolic autocorrelation. The temporal parameters are identified, yet the spatial hold-out score barely moves: with every day of the record observed at the training sites, predicting a withheld site is an interpolation problem, and the daily cross-sectional information dominates whatever the autoregressive order contributes.

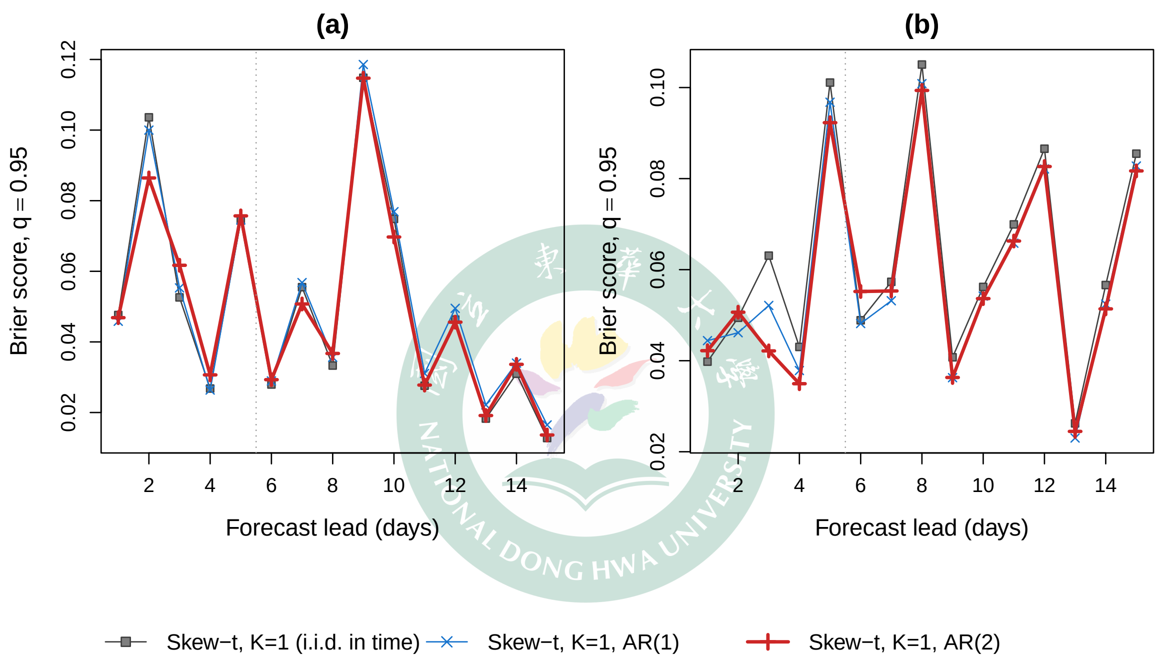



Figure 3: Time-block forecast arm of the persistence experiment: mean Brier score at the $0.95$ threshold quantile by forecast lead (10 data sets; lower is better; the dotted vertical line marks the lead $1$–$5$ window). Data-generating settings: (a) near-unit-root AR(2), $(\phi_1, \phi_2) = (0.15, 0.80)$; (b) long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$.

The time-block forecast arm asks the question the spatial split cannot: whether the fitted temporal structure improves genuine forward-in-time prediction (Figure 3, Table 3). Under long memory the answer is affirmative. The AR(2) model attains the lowest Brier score at $12$ of the $15$ leads, improving on the i.i.d.-in-time fit by about $12\%$ over leads $1$–$5$ ($0.0524$ versus $0.0593$) and by about $7\%$ over the full horizon, with the AR(1) model in between; this ordering is an order of magnitude larger than any difference the spatial arm produced.

Under the near-unit-root design the per-lead profile is noisier (each lead contributes a single binary exceedance event per site and day) and the three models are close over the first five leads; pooled over all $15$ leads the AR(2) fit is nevertheless the best of the three ($0.0495$, against $0.0498$ for i.i.d. and $0.0515$ for AR(1)). With $10$ data sets none of the paired Brier differences reaches statistical significance, so these results should be read as directional: the gains are modest, but they appear exactly where the structure of the task predicts them, namely in temporal extrapolation under strong persistence, and they are largest in the regime where the second lag improves the finite-order approximation.

Table 3: Time-block forecast arm: mean Brier score at the $0.95$ threshold quantile by lead window ($10$ data sets; lower is better). The fitted models are the skew-$t$ $K = 1$ variants with i.i.d., AR(1), and AR(2) latent temporal structure.

| Generating design | Leads | i.i.d. | AR(1) | AR(2) | AR(2)−i.i.d. | AR(2)−AR(1) |
|---|---|---|---|---|---|---|
| Near-unit-root AR(2) | 1–5 | 0.0610 | 0.0604 | 0.0603 | −0.0007 | −0.0001 |
| Near-unit-root AR(2) | 1–15 | 0.0498 | 0.0515 | 0.0495 | −0.0003 | −0.0020 |
| ARFIMA, $d = 0.45$ | 1–5 | 0.0593 | 0.0555 | 0.0524 | −0.0068 | −0.0030 |
| ARFIMA, $d = 0.45$ | 1–15 | 0.0620 | 0.0584 | 0.0579 | −0.0041 | −0.0005 |

### 3.5 Results on MRTS comparison

On the two skew-$t$ -core designs, augmenting the regression component with MRTS basis functions does not deliver a stable improvement in upper-tail scores (Figure 4). As with the AR(2) comparison, the data-generating knot count dominates the ranking: the matching five-knot skew-$t$ model is best on the five-knot setting, and the one-knot model is best on the random-effect process.

Enlarging the MRTS basis does not lower the mean Brier score on either setting. On the five-knot setting every curve is essentially flat in $K_{\text{MRTS}}$. On the additive non-stationary spatial random-effect process the curves are likewise flat, with over-specified models destabilising at large $K_{\text{MRTS}}$. This second setting is the decisive one: the non-stationarity lives in the second moment (a daily-redrawn random effect $u_t$), whereas the MRTS basis enters only the fixed-effect mean. Because $u_t$ has zero time-average, the time-pooled coefficient $\boldsymbol{\beta}^*$ has no first-moment structure to recover, so the Brier- and recovery-versus-$K$ curves stay flat by construction, not for want of signal, since $u_t$ contributes substantial (spatially non-stationary) variance. The MRTS mean term therefore yields no reliable tail-score gain: a fixed-effect mean cannot represent a non-stationary second-moment feature, and so the spatial mean is not the binding constraint on upper-tail prediction.

The two nonlinear-mean designs probe the complementary regime (Figure 5; Table 4). On the arc-front design the Brier score finally responds to the mean basis, but only once the basis is fine enough to resolve the front. At the $0.95$ threshold quantile the profiles are flat up to $K_{\mathrm{MRTS}} = 8$, deteriorate by 2–4% around $K_{\mathrm{MRTS}} = 10$–$12$, and only then improve, reaching their minima at $K_{\mathrm{MRTS}} = 25$–$50$ with gains of 3–4% for the unthresholded models and 5–6% for the thresholded ones (8–9% and 6%, respectively, at the $0.98$ quantile). The mid-$K$ dip is the signature of a too-coarse basis: a partially fitted mean sharpens the predictive distribution while leaving a residual surface error that extrapolates to the held-out sites, and the net effect on the Brier score is negative until the basis actually resolves the front.

The transform-mix pilot shows the same mechanism operating from a much worse starting point. The mismatched five-knot partition (compounded, for the thresholded variant, by censoring) leaves the $K_{\mathrm{MRTS}} = 0$ baselines of the two five-knot models ($0.0199$ and $0.0119$ at the $0.95$ quantile) far above those of the one-knot models ($0.0067$–$0.0085$), and the mean basis buys most of that damage back: every model improves at the $0.95$ quantile (relative Brier $0.20$–$0.64$), and at its best basis size each model converges to nearly the same absolute score (about $0.004$). The rescue is not uniform across levels, however: at the $0.98$ quantile the three one-knot fits come out 18–23% *worse* than their $K_{\mathrm{MRTS}} = 0$ baselines, a reminder that with $3$ data sets and exceedances this rare the pilot profiles are indicative only. Both surfaces were chosen, by an oracle calculation performed before any fitting, to leave genuine Brier headroom: replacing the fitted mean by the true surface $g$ under the true Gaussian error law caps the level-averaged relative Brier score at $0.56$ (arc front) and $0.40$ (transform mix). The realized gains convert only a small part of that headroom, which is consistent with the sharpness–extrapolation account above: the Brier score improves only where the residual basis error is small relative to the predictive spread.

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the STP baseline. The AR(2) prior provides temporal flexibility and the MRTS covariates enrich the spatial regression component, yet their empirical benefits depend on the data-generating mechanism and on the target exceedance level. The persistence experiment sharpens this conclusion into a statement about the prediction task itself: even under near-unit-root and long-memory dependence, with the temporal coefficients well identified, spatial hold-out prediction is insensitive to the autoregressive order, whereas modest but consistent gains from richer temporal pooling emerge once the task requires ex-

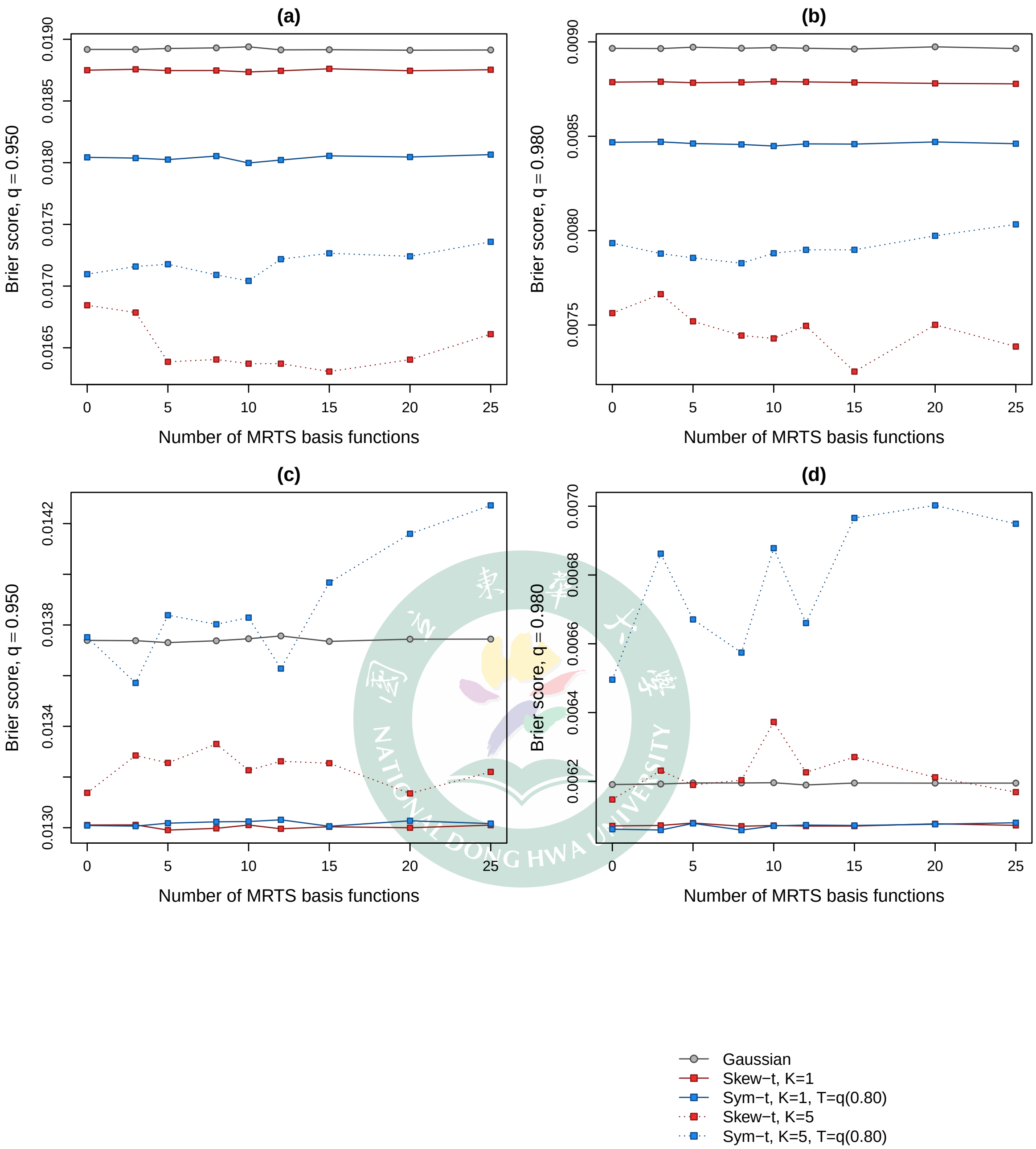



Figure 4: Brier score versus the number of MRTS basis functions $K_{\text{MRTS}}$ ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) skew-$t$ ($K = 5$, $\lambda = 3$); (c,d) skew-$t$ ($K = 1$, $\lambda = 3$) with an additive non-stationary spatial random effect. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

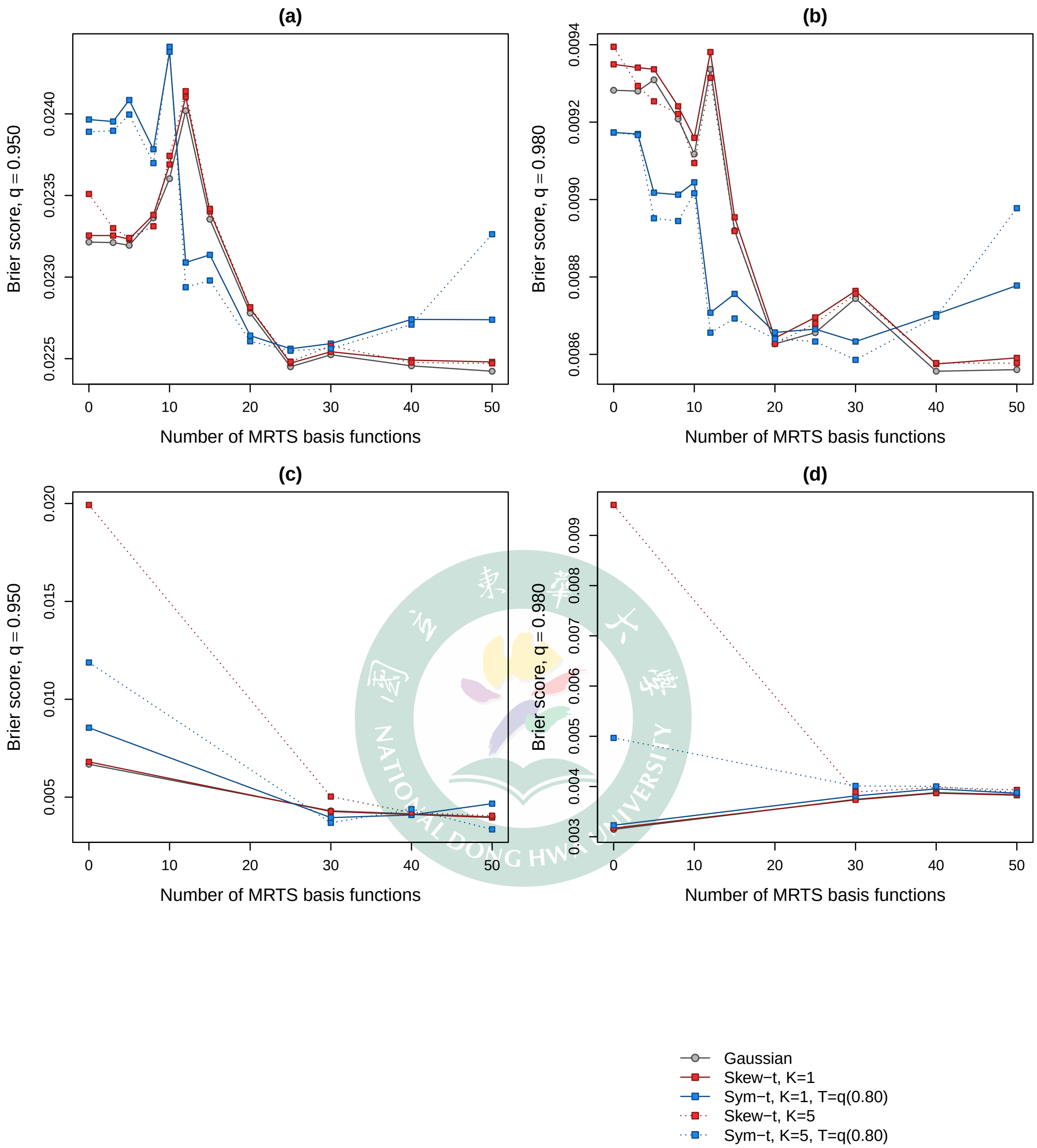



Figure 5: Brier score versus the number of MRTS basis functions $K_{\text{MRTS}}$ on the two nonlinear-mean designs of (29) ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) arc front ($10$ data sets, full $K_{\text{MRTS}}$ grid); (c,d) transform mix ($3$-data-set pilot, fitted at $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$). Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

Table 4: Nonlinear-mean designs: Brier score at the $0.95$ threshold quantile relative to the same model's own $K_{\text{MRTS}} = 0$ fit (lower is better). Best is the smallest ratio over $K_{\text{MRTS}} > 0$, with the minimizing basis size in parentheses; $\text{BS}_0$ is the absolute Brier score at $K_{\text{MRTS}} = 0$; thr. denotes the POT threshold $T = q(0.80)$. The arc-front design uses $10$ data sets on the full grid; the transform-mix design is a 3-data-set pilot fitted at $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$.

| Generating design | Analysis model | $K$=30 | $K$=40 | $K$=50 | Best ($K$) | $\text{BS}_0$ |
|---|---|---|---|---|---|---|
| Arc front | Gaussian | 0.970 | 0.967 | 0.966 | 0.966 (50) | 0.0232 |
| Arc front | Skew-$t$ , $K = 1$ | 0.969 | 0.967 | 0.967 | 0.966 (25) | 0.0233 |
| Arc front | Symmetric-$t$, $K = 1$, thr. | 0.943 | 0.949 | 0.949 | 0.941 (25) | 0.0240 |
| Arc front | Skew-$t$ , $K = 5$ | 0.960 | 0.956 | 0.956 | 0.956 (50) | 0.0235 |
| Arc front | Symmetric-$t$, $K = 5$, thr. | 0.944 | 0.951 | 0.974 | 0.944 (25) | 0.0239 |
| Transform mix | Gaussian | 0.644 | 0.621 | 0.597 | 0.597 (50) | 0.0067 |
| Transform mix | Skew-$t$ , $K = 1$ | 0.628 | 0.605 | 0.582 | 0.582 (50) | 0.0068 |
| Transform mix | Symmetric-$t$, $K = 1$, thr. | 0.463 | 0.479 | 0.546 | 0.463 (30) | 0.0085 |
| Transform mix | Skew-$t$ , $K = 5$ | 0.252 | 0.211 | 0.203 | 0.203 (50) | 0.0199 |
| Transform mix | Symmetric-$t$, $K = 5$, thr. | 0.311 | 0.369 | 0.282 | 0.282 (50) | 0.0119 |

trapolating beyond the observed record. The AR(2) extension is therefore best regarded as a forecasting device rather than an interpolation device. The MRTS experiments admit an equally structural summary: the mean term is inert when the non-stationarity lives in the second moment, and under genuine nonlinear mean structure it pays off only once the basis is fine enough to resolve the surface, converting a small fraction of the oracle headroom. Whether the same pattern holds on real environmental data (and, if so, what it implies for practitioners choosing among the candidate settings) is the question taken up in the Data analysis section, where AR(2) and MRTS variants are treated as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) STP baseline.

# 4 Real data analysis

Morris, Reich, Emeric Thibaud, and Cooley (2017) introduced the spatio-temporal skew-$t$ (STP) process to predict high-threshold ground-level ozone exceedances from U.S. EPA monitoring data. We return to that same data here, asking not for a new model class but whether enlarging the STP candidate class along the two axes developed in this thesis, an AR(2) temporal prior and an MRTS spatial-mean term, yields better held-out tail prediction. The comparison is therefore designed to identify the exceedance levels, if any, at which the enlarged candidate class improves on the STP baseline, rather than to select a single overall-best model.

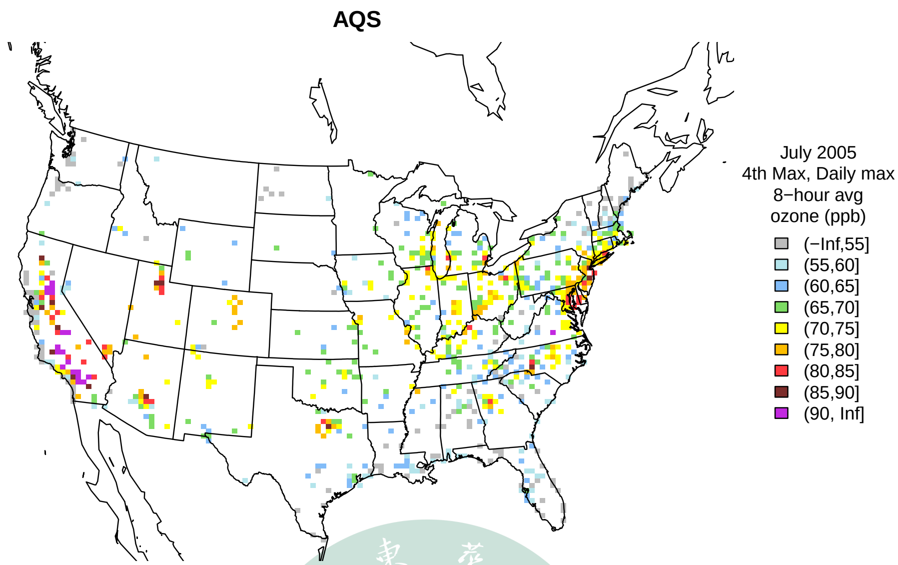
The response is the daily maximum 8-hour ozone concentration for the $31$ days of July 2005 at United States Air Quality System (AQS) monitoring sites, with the co-located Community Multi-scale Air Quality (CMAQ) model output as the sole covariate. Figure 6 summarizes the two data sources through the EPA design value, the fourth-highest daily maximum 8-hour average of the month. The AQS monitors are irregularly spaced, covering the eastern United States and California densely but leaving large gaps in the interior west, whereas CMAQ provides complete gridded coverage of the conterminous United States. CMAQ reproduces the broad spatial pattern of the observations but not their level at individual monitors, which motivates using it as a regression covariate to be calibrated rather than as a substitute for the observations. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), the regression component is $\mathbf{X}_t(\mathbf{s}) = [1, \mathrm{CMAQ}_t(\mathbf{s})]^\top$, so that each model acts as a statistical calibration of CMAQ against the monitor observations.

The distributional features of these data are what motivate the skew-$t$ marginal in the first place. Regressing the observations on CMAQ at the $1{,}089$ sites above and pooling the residuals across sites and days, the QQ plot against the Normal distribution departs from the diagonal in both tails, with the upper tail by far the heavier (Figure 7); the largest standardized residuals reach eight standard deviations, far beyond what a Gaussian model can accommodate. A skew-$t$ distribution fit to the pooled residuals by maximum likelihood, with slant $\hat{\alpha} = 0.23$ and $\hat{\nu} = 15.7$ degrees of freedom, straightens the plot except for a handful of extreme points. The residual ozone variation is thus heavy-tailed and right-skewed exactly in the region that exceedance prediction cares about, and a skew-$t$ process is a natural candidate class for it.

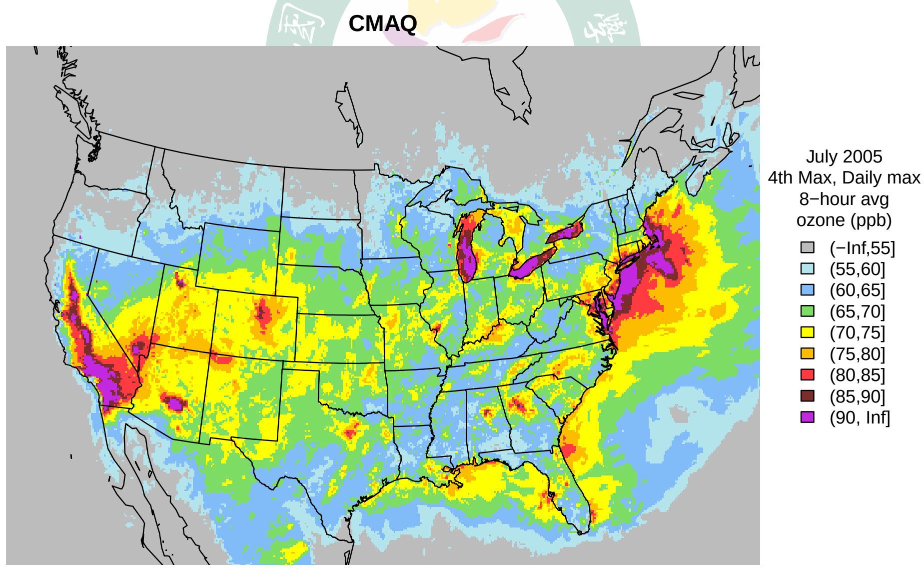
## 4.1 Model comparisons

The baseline candidate set follows that of Morris, Reich, Emeric Thibaud, and Cooley (2017). We fit Gaussian and skew-$t$ marginals over a range of partition sizes, $K \in \{1, 5, 6, 7, 8, 9, 10, 15\}$, and censor the response at three levels to contrast no, moderate, and high censoring: no censoring ($T = 0$), a moderate threshold ($T = 50$ ppb, the $0.42$ sample quantile), and a high threshold ($T = 75$ ppb, the $0.92$ sample quantile). For settings censored at $T = 75$ ppb, we use a symmetric-$t$ marginal, as in the simulation study. Each setting is fit both with and without a latent time series.

On top of this baseline we add the two extensions developed in this thesis. The first replaces the AR(1) latent dynamics with the standardized AR(2) prior of Section 2.3; the second augments the regression component with MRTS basis functions, so that non-stationary spatial mean structure can be absorbed by multi-resolution thin-plate splines.




(a) AQS monitor observations




(b) CMAQ model output

Figure 6: July 2005 ozone design value (fourth-highest daily maximum 8-hour average, in ppb): (a) at the 1,089 AQS monitoring sites missing at most 50% of the 31 days; (b) on the full CMAQ grid.

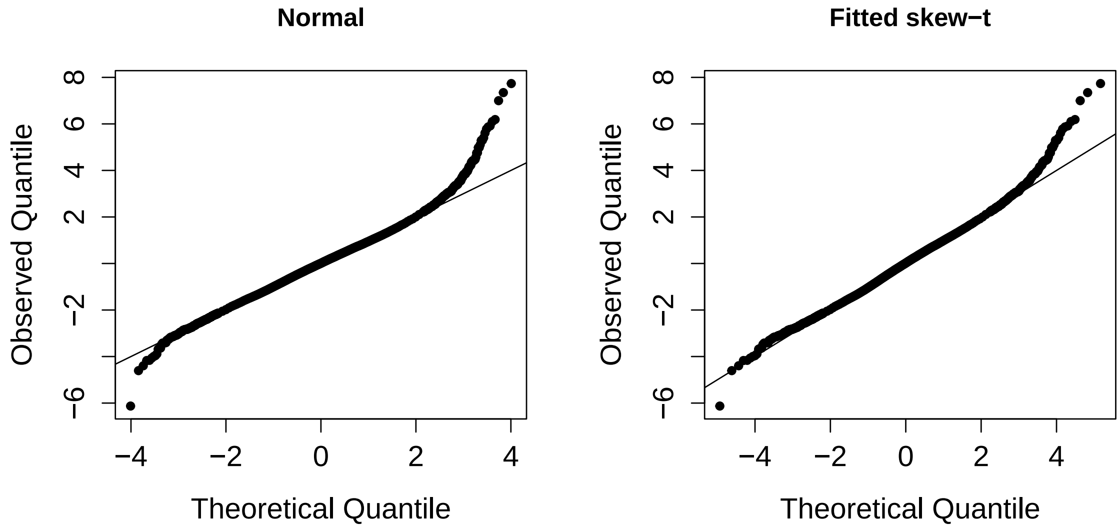



Figure 7: QQ plots of the pooled residuals from the linear regression of ozone on CMAQ at the $1{,}089$ AQS sites, against the Normal distribution (left) and the ML-fitted skew-$t$ distribution with $\hat{\alpha} = 0.23$ and $\hat{\nu} = 15.7$ (right).

Each MCMC chain is run for $30{,}000$ iterations with a burn-in period of $25{,}000$ iterations. The comparison draws on $800$ monitoring sites and is evaluated by two-fold cross validation, with $400$ training and $400$ validation sites per fold, scored by the held-out Brier score of Section 3.2.

## 4.2 Results

Across the upper tail, the proposed extensions surface among the best models at scattered exceedance levels rather than improving prediction uniformly, so no single extension, and no single setting, is best across the whole tail (Table 5). The extensions nonetheless earn their place by supplying the leading model at specific points along the tail that the baseline alone does not reach.

This picture is reproducible on smaller training samples. Repeating the comparison under two further cross-validation splits, written $N_{\text{train}}/N_{\text{test}}$ for $N_{\text{train}}$ training and $N_{\text{test}}$ validation sites (the 300/100 and 200/200 splits; Tables 6 and 7), recovers the same qualitative ranking: the Baseline, AR(2), and MRTS families continue to share the top ranks across the tail, even as the exact best-ranked setting shifts with the split and the level. That shifting is expected, since only a handful of validation sites exceed the highest thresholds and the fine ranking

Table 5: Top two performing models for predicting ozone exceedance of level $L$ with Brier score.

| Level $L$ | Rank 1 model | Rank 2 model |
|---|---|---|
| 0.900 | **MRTS** skew-$t$ , $K = 1$, $T = 0$, AR(1), $K_{\mathrm{MRTS}} = 10$ | skew-$t$ , $K = 1$, $T = 0$, AR(1) |
| 0.950 | **AR2** skew-$t$ , $K = 7$, $T = 50$ | **AR2** skew-$t$ , $K = 10$, $T = 50$ |
| 0.980 | skew-$t$ , $K = 6$, $T = 50$, AR(1) | skew-$t$ , $K = 5$, $T = 50$, AR(1) |
| 0.990 | symmetric-$t$, $K = 9$, $T = 75$, AR(1) | skew-$t$ , $K = 5$, $T = 50$ |
| 0.995 | **AR2** symmetric-$t$, $K = 15$, $T = 75$ | symmetric-$t$, $K = 9$, $T = 75$, AR(1) |

Table 6: Top four Brier score models for the 300/100 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

| Level $L$ | Rank 1 | Rank 2 | Rank 3 | Rank 4 |
|---|---|---|---|---|
| 0.900 | **MRTS** (0.041) | Baseline (0.041) | **MRTS** (0.041) | Baseline (0.041) |
| 0.950 | Baseline (0.020) | Baseline (0.020) | Baseline (0.020) | Baseline (0.020) |
| 0.980 | **MRTS** (0.008) | **AR2** (0.008) | Baseline (0.008) | **AR2** (0.008) |
| 0.990 | **AR2** (0.003) | Baseline (0.003) | Baseline (0.003) | Baseline (0.003) |
| 0.995 | Baseline (0.001) | Baseline (0.001) | Baseline (0.001) | **AR2** (0.001) |

Table 7: Top four Brier score models for the 200/200 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions.

| Level $L$ | Rank 1 | Rank 2 | Rank 3 | Rank 4 |
|---|---|---|---|---|
| 0.900 | Baseline (0.050) | **AR2** (0.050) | Baseline (0.050) | **MRTS** (0.050) |
| 0.950 | **AR2** (0.030) | **AR2** (0.030) | **AR2** (0.030) | Baseline (0.030) |
| 0.980 | **MRTS** (0.015) | **MRTS** (0.015) | **AR2** (0.015) | Baseline (0.015) |
| 0.990 | Baseline (0.008) | **AR2** (0.008) | **AR2** (0.008) | Baseline (0.008) |
| 0.995 | Baseline (0.004) | Baseline (0.004) | Baseline (0.004) | **AR2** (0.004) |

there is sensitive to the held-out draw; the persistence of the broad pattern across splits is what supports retaining AR(2) and MRTS as candidates.

# 5 Discussion

This thesis asked not whether the spatio-temporal skew-$t$ process predicts high-threshold exceedances well, which Morris, Reich, Emeric Thibaud, and Cooley (2017) already established, but where within its four components room for sharper upper-tail prediction remains. To probe it, we enlarged the model along two opposing axes: an AR(2) latent prior adding temporal memory, and an MRTS mean term adding a flexible non-stationary mean (a spatial trend). We scored both on held-out data rather than asserting either as a default. The temporal axis yielded almost no usable signal: even on AR(2)-generated data the temporal curves sat on top of their non-temporal counterparts at every quantile, and the recovered coefficients (Table 2) were only weakly identified: recovery was reliable mainly for the scale latent under strong dependence at matched low-$K$ settings, while recovery for the skewness and knot-location latents was unstable, with large posterior standard deviations and attenuation toward zero, especially as $K$ increased or dependence weakened. Tail prediction is set by how faithfully the marginal and the spatial partition capture the upper tail, not by the memory of the latent coefficients. The spatial axis was more favorable but no more stable, with the MRTS basis improving selected scores yet failing to hold across families or quantile levels; and while the far-tail ranking is sensitive to the handful of sites exceeding the highest thresholds, so that the single best setting shifts with the split, the broad pattern in which the baseline, AR(2), and MRTS families recur among the top ranks persists across all splits, and it is that persistence that justifies retaining the extensions.

The lesson reaches past ozone, since the same target, a spatial probability statement about the upper tail at unmonitored sites, arises for any heavy-tailed, asymmetric environmental field regulated by a high threshold, from precipitation extremes to heat warnings: temporal order and mean flexibility are not free improvements to such a model, but neither are they idle. Rather than replacing the STP baseline, the standardized AR(2) prior and the MRTS mean enlarge it into a candidate class from which held-out tail scores can select the setting that fits the data at hand, giving practitioners who must predict high-threshold exceedances a broader, data-driven choice than any single fixed model can offer.

# A Appendices

## A.1 Stationary moments of the standardized AR(2)

We derive the Yule–Walker moments (13)–(14) used to standardize the AR(2) prior (11) to unit marginal variance. Let $\gamma_h = \mathrm{Cov}(Y_t, Y_{t-h})$ be the stationary autocovariances, normalized so that $\gamma_0 = \mathrm{Var}(Y_t) = 1$. Multiplying the recursion (11) by $Y_{t-h}$ for $h \geq 1$ and taking expectations, and using that $\epsilon_t$ is independent of $Y_{t-h}$, yields the Yule–Walker recursion

$$\gamma_h \; = \; \phi_1 \, \gamma_{h-1} \; + \; \phi_2 \, \gamma_{h-2}, \qquad h \geq 1. \tag{31}$$

Evaluating (31) at $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ with $\gamma_0 = 1$ gives $\gamma_1 = \phi_1 + \phi_2 \, \gamma_1$, hence $\gamma_1 = \phi_1/(1 - \phi_2)$, and at $h = 2$ gives $\gamma_2 = \phi_1 \, \gamma_1 + \phi_2$; together these are (13). Multiplying (11) instead by $Y_t$ and taking expectations, with $\mathrm{E}[Y_t \, \epsilon_t] = \sigma_\epsilon^2$ because $Y_t$ carries the current innovation, gives $\gamma_0 = \phi_1 \, \gamma_1 + \phi_2 \, \gamma_2 + \sigma_\epsilon^2$, and solving for $\sigma_\epsilon^2$ under $\gamma_0 = 1$ recovers (14). The same normalization fixes the initial pair: $(Y_1, Y_2)$ has unit variances and lag-1 covariance $\gamma_1$, which is the stationary bivariate Gaussian (15).

## A.2 AR(2) full conditional for the Gibbs-within-MH update

The AR(2) Markov structure factorizes the joint prior of $(Y_1, \ldots, Y_{n_t})$ as

$$p(Y_1, \ldots, Y_{n_t}) \; = \; p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \tag{32}$$

with $p(Y_1, Y_2)$ given by (15) and each conditional factor for $s \geq 3$ following (11). For an interior time point $3 \leq t \leq n_t - 2$, the value $Y_t$ enters (32) through three factors: the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of $Y_t$ is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \; \propto \; L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \tag{33}$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by $Y_t$ through the partition and Gaussian process structure of (1). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim$

$N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R \;=\; \min\!\left(1,\; \frac{\pi(Y_t^{(c)} \mid Y_{-t}, \mathbf{Y})}{\pi(Y_t \mid Y_{-t}, \mathbf{Y})}\right), \tag{34}$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (13).

### A.3 Marginal covariance of the non-stationary-dependence design

The second MRTS design of Section 3.1 superposes two fixed cosine fields (26) with daily-redrawn Gaussian weights $\xi_{tj} \overset{iid}{\sim} N(0, \tau_j^2)$ (27). Marginalising over the weights, $u_t(\mathbf{s}) = \sum_j \xi_{tj}\, \phi_j(\mathbf{s})$ has zero mean, $\mathrm{E}[u_t(\mathbf{s})] = \sum_j \phi_j(\mathbf{s})\, \mathrm{E}[\xi_{tj}] = 0$, and, by the independence of $\xi_{t1}$ and $\xi_{t2}$, the low-rank covariance

$$\mathrm{Cov}\big(u_t(\mathbf{s}), u_t(\mathbf{s}')\big) \;=\; \sum_{j=1}^{2} \tau_j^2\, \phi_j(\mathbf{s})\, \phi_j(\mathbf{s}'). \tag{35}$$

The pointwise variance $\mathrm{Var}\big(u_t(\mathbf{s})\big) = \sum_j \tau_j^2\, \phi_j(\mathbf{s})^2$ depends on location, so the dependence is non-stationary while the marginal mean stays constant at 10.



## References

Coles, Stuart, Janet Heffernan, and Jonathan Tawn (1999). "Dependence Measures for Extreme Value Analyses". In: *Extremes* 2.4, pp. 339–365. ISSN: 1572-915X. DOI: 10.1023/A:1009963131610. URL: https://doi.org/10.1023/A:1009963131610.

Davies, R. B. and D. S. Harte (1987). "Tests for Hurst effect". In: *Biometrika* 74.1, pp. 95–101. DOI: 10.1093/biomet/74.1.95.

Davison, Anthony C. and R. L. Smith (1990). "Models for exceedances over high thresholds (with Discussion)". In: *Journal of the Royal Statistical Society. Series B (Methodological)* 52.3, pp. 393–442. ISSN: 00359246.

Gneiting, Tilmann and Adrian E Raftery (Mar. 2007). "Strictly Proper Scoring Rules, Prediction, and Estimation". In: *Journal of the American Statistical Association* 102.477. _eprint: https://doi.org/10.1198/016214506000001437, pp. 359–378. ISSN: 0162-1459. DOI:

10.1198/016214506000001437. URL: https://doi.org/10.1198/016214506000001437 (visited on 11/10/2025).

Granger, C. W. J. and Roselyne Joyeux (1980). “An Introduction to Long-Memory Time Series Models and Fractional Differencing”. In: *Journal of Time Series Analysis* 1.1, pp. 15–29. DOI: 10.1111/j.1467-9892.1980.tb00297.x.

Hosking, J. R. M. (1981). “Fractional Differencing”. In: *Biometrika* 68.1, pp. 165–176. DOI: 10.1093/biomet/68.1.165.

Huser, R. and A. C. Davison (Mar. 2014). “Space-time modelling of extreme events”. In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 76.2, pp. 439–461. ISSN: 13697412. arXiv: 1201.3245.

Huser, Raphaël and Jennifer L. Wadsworth (2022). “Advances in statistical modeling of spatial extremes”. en. In: *WIREs Computational Statistics* 14.1. _eprint: https://wires.onlinelibrary.wiley.com/do e1537. ISSN: 1939-0068. DOI: 10.1002/wics.1537. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/wics.1537 (visited on 07/02/2026).

Ledford, Anthony W. and Jonathan A. Tawn (1996). “Statistics for near independence in multivariate extreme values.” In: *Biometrika* 83, pp. 169–187. ISSN: 0006-3444.

Morris, Samuel A., Brian J. Reich, and Emeric Thibaud (Dec. 2019). “Exploration and Inference in Spatial Extremes Using Empirical Basis Functions”. en. In: *JABES* 24.4, pp. 555–572. ISSN: 1537-2693. DOI: 10.1007/s13253-019-00359-1. URL: https://doi.org/10.1007/s13253-019-00359-1 (visited on 02/10/2026).

Morris, Samuel A., Brian J. Reich, Emeric Thibaud, and Daniel Cooley (Sept. 2017). “A Space-Time Skew-t Model for Threshold Exceedances”. In: *Biometrics* 73.3, pp. 749–758. ISSN: 0006-341X. DOI: 10.1111/biom.12644. URL: https://doi.org/10.1111/biom.12644 (visited on 03/14/2025).

Padoan, S. A., M. Ribatet, and S. A. Sisson (Mar. 2010). “Likelihood-Based Inference for Max-Stable Processes”. In: *Journal of the American Statistical Association* 105.489, pp. 263–277. ISSN: 0162-1459.

Thibaud, E., R. Mutzner, and A. C. Davison (2013). “Threshold modeling of extreme spatial rainfall”. In: *Water Resources Research* 49.8, pp. 4633–4644. ISSN: 00431397.

Tzeng, ShengLi and Hsin-Cheng Huang (Apr. 2018). “Resolution Adaptive Fixed Rank Kriging”. In: *Technometrics* 60.2. _eprint: https://doi.org/10.1080/00401706.2017.1345701, pp. 198–208. ISSN: 0040-1706. DOI: 10.1080/00401706.2017.1345701. URL: https://doi.org/10.1080/00401706.2017.1345701 (visited on 03/22/2026).

U.S. Environmental Protection Agency (Oct. 2024). *Bayesian Space-time Downscaling Fusion Model (Downscaler) Derived Estimates of Air Quality for 2021*. Annual Report EPA-454/R-24-002. Office of Air Quality Planning and Standards, U.S. EPA. URL: https://www.epa.gov/hesc/rsig-related-downloadable-data-files.

Wadsworth, Jennifer L. and Jonathan A. Tawn (2012). "Dependence modelling for spatial extremes". In: *Biometrika* 99.2, pp. 253–272. ISSN: 00063444.